

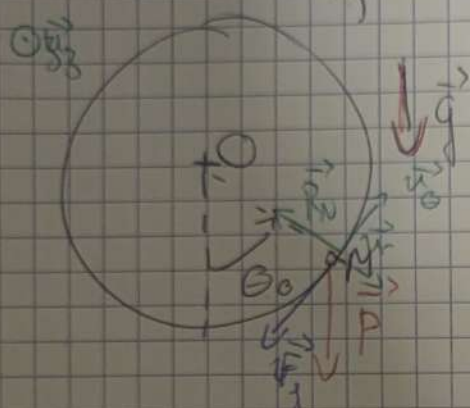
Ex 6.

1/ Système: Une articule M, de masse m.

Ref terrestre Supposé galiléen

mouvement circulaire de rotation axé (Oz) $\Rightarrow$ Base cylindrique $(\vec{u}_r, \vec{u}_\theta, \vec{u}_z)$

Bilan des forces:



$$\begin{aligned} \vec{F}_p &= -mg \vec{u}_z \\ \vec{P} &= m\vec{g} \\ \vec{R}_N &= R_N \vec{u}_z \quad \text{avec } R_N > 0 \end{aligned}$$

Cinématique

$$\begin{aligned} \vec{OM} &= R \vec{u}_r \\ \vec{v} &= R \dot{\theta} \vec{u}_\theta \end{aligned}$$

Projection des forces sur la base $(\vec{u}_r, \vec{u}_\theta, \vec{u}_z)$

$$\vec{P} = m \begin{cases} \vec{g} \cdot \vec{u}_r = g \cos(\theta_0 + \frac{\pi}{2}) = -g \sin(\theta_0) \\ \vec{g} \cdot \vec{u}_\theta = g \cos(\theta_0) = g \sin(\theta_0) \end{cases}$$

$$\vec{F}_f = -R \dot{\theta} \vec{u}_\theta$$

Moment de $\vec{P}$ p/r à O:

$$\vec{M}_O(\vec{P}) = \vec{OM} \wedge \vec{P}$$

$$= -Rmg \sin(\theta_0) \vec{u}_z$$

Moment de $\vec{F}_f$ p/r à O:

$$\vec{M}_O(\vec{F}_f) = \vec{OM} \wedge \vec{F}_f$$

$$\begin{aligned}\vec{M}_O(\vec{F}) &= \vec{OM} \wedge \vec{F} \\ &= R\vec{u}_r \wedge (-\alpha R\dot{\theta}\vec{u}_\theta) \\ &= -\alpha R^2\dot{\theta}\vec{u}_z\end{aligned}$$

$D(R, \vec{R}_N)$ passe par O donc $\vec{M}_O(\vec{R}_N) = 0$

Moment cinétique:

$$\begin{aligned}\vec{L}_O(\vec{M}) &= \vec{OM} \wedge m\vec{v} \\ &= R\vec{u}_r \wedge mR\dot{\theta}\vec{u}_\theta \\ &= mR^2\dot{\theta}\vec{u}_z\end{aligned}$$

Moment cinétique scalaire:

$$\begin{aligned}L_{(Oz)}(M) &= \vec{L}_O(M) \cdot \vec{u}_z \\ &= mR^2\dot{\theta}\end{aligned}$$

Moment scalaire de $\vec{P}$:

$$\begin{aligned}M_{(Oz)}(\vec{P}) &= \vec{M}_{(Oz)}(\vec{P}) \cdot \vec{u}_z \\ &= -Rmg \sin(\theta)\end{aligned}$$

Moment scalaire de $\vec{F}$:

$$\begin{aligned}M_{(Oz)}(\vec{F}) &= \vec{M}_{(Oz)}(\vec{F}) \cdot \vec{u}_z \\ &= -\alpha R^2\dot{\theta}\end{aligned}$$

Théorème du moment cinétique scalaire:

$$\frac{dL_{(Oz)}(M)}{dt} = M_{(Oz)}(\vec{P}) + M_{(Oz)}(\vec{F})$$

$$\Rightarrow mR^2\ddot{\theta} = -Rmg \sin(\theta) - \alpha R^2\dot{\theta}$$

$$\Rightarrow mR^2\ddot{\theta} + \alpha R^2\dot{\theta} + Rmg \sin(\theta) = 0 \Leftrightarrow \ddot{\theta} + \frac{\alpha}{m}\dot{\theta} + \frac{g}{R}\sin(\theta) = 0$$

Approximation des angles : θ très petit $\Rightarrow \sin(\theta) \approx \theta$

$$\Rightarrow \ddot{\theta} + \frac{g}{l} \theta + \frac{1}{R} \theta = 0$$

$$\Rightarrow \ddot{\theta} + 2\zeta \omega_0 \dot{\theta} + \omega_0^2 \theta = 0$$

avec $\omega_0 = \sqrt{\frac{g}{l}}$

$$2\zeta \omega_0 = \frac{1}{m}$$

$$\Rightarrow \zeta = \frac{1}{m} \times \frac{1}{2\omega_0}$$

$$\zeta = \frac{1}{2m} \times \sqrt{\frac{R}{g}}$$

$$b' = \frac{b}{2}$$

2). Eq carac: $x^2 + 2\zeta \omega_0 x + \omega_0^2$

Delta: $\Delta = \frac{b'^2}{a^2} - \frac{c}{a} = \zeta^2 \omega_0^2 - \omega_0^2 = \omega_0^2 (\zeta^2 - 1)$

$$\sqrt{\Delta} = \omega_0 \sqrt{\zeta^2 - 1}$$

Cas critique: $\zeta = 1$

$$\Rightarrow \frac{1}{2m} \sqrt{\frac{R}{g}} = 1$$

$$\Rightarrow \frac{1}{4m^2} \frac{R}{g} = 1$$

$$\Rightarrow R_c = \frac{4m^2 g}{1^2}$$

Résolution de l'éq. diff:

Racines: $\frac{-b'}{2a} = \frac{-b'}{a} = -\zeta \omega_0$

Solution de l'exercice

$$\Theta(t) = (A + Bt) e^{-\zeta \omega_0 t}$$

CI: $\Theta(0) = \Theta_0$

$$\dot{\Theta}(0) = 0$$

multiplier: $\Theta(0) = A$
 $\Rightarrow A = \Theta_0$

$$\dot{\Theta}(t) = B e^{-\zeta \omega_0 t} - \zeta \omega_0 (A + Bt) e^{-\zeta \omega_0 t}$$

$$\Rightarrow \dot{\Theta}(0) = B - \zeta \omega_0 A$$

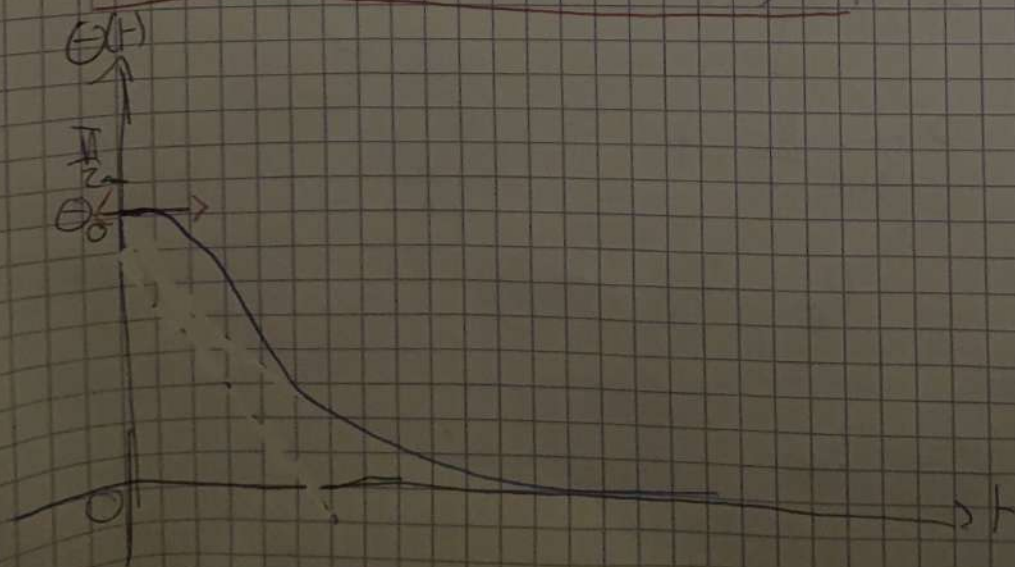
$$B = \zeta \omega_0 A$$

$$B = \zeta \omega_0 \Theta_0$$

Solution finale:

$$\Theta(t) = \Theta_0 (1 + \zeta \omega_0 t) e^{-\zeta \omega_0 t}$$

$$\Theta(t) = \Theta_0 (1 + \omega_0 t) e^{-\omega_0 t} \quad \text{car } \zeta = 1$$



$$3) \dot{\Theta}(t) = \Theta_0 \omega_0 e^{-\omega_0 t} - \omega_0 \Theta_0 (1 + \omega_0 t) e^{-\omega_0 t}$$

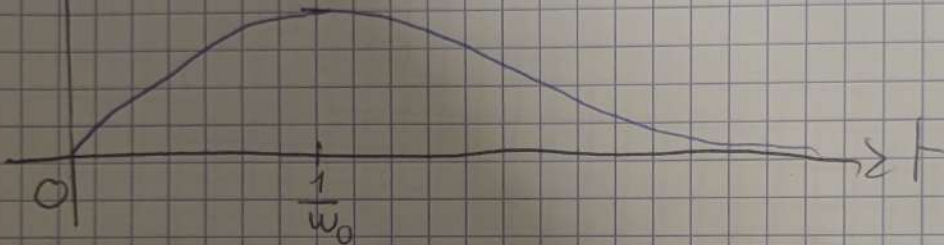
$$= \Theta_0 \omega_0 (1 - 1 - \omega_0 t) e^{-\omega_0 t}$$

$$\dot{\Theta}(t) = -\Theta_0 \omega_0^2 t e^{-\omega_0 t}$$

$$\Rightarrow \vec{v}(t) = R_c \dot{\Theta}(t) \vec{u}_\theta$$

$$\Rightarrow \vec{v}(t) = -R_c \Theta_0 \omega_0^2 t e^{-\omega_0 t}$$

$$\|\vec{v}\| = R_c \dot{\Theta}(t)$$



~~$$\ddot{\Theta}(t) = -\Theta_0 \omega_0^2 (e^{-\omega_0 t} - \omega_0 t e^{-\omega_0 t})$$

$$= -\Theta_0 \omega_0^2 (1 - \omega_0 t) e^{-\omega_0 t}$$

$$\ddot{\Theta} = \frac{1}{\omega_0} \ddot{\Theta}$$~~